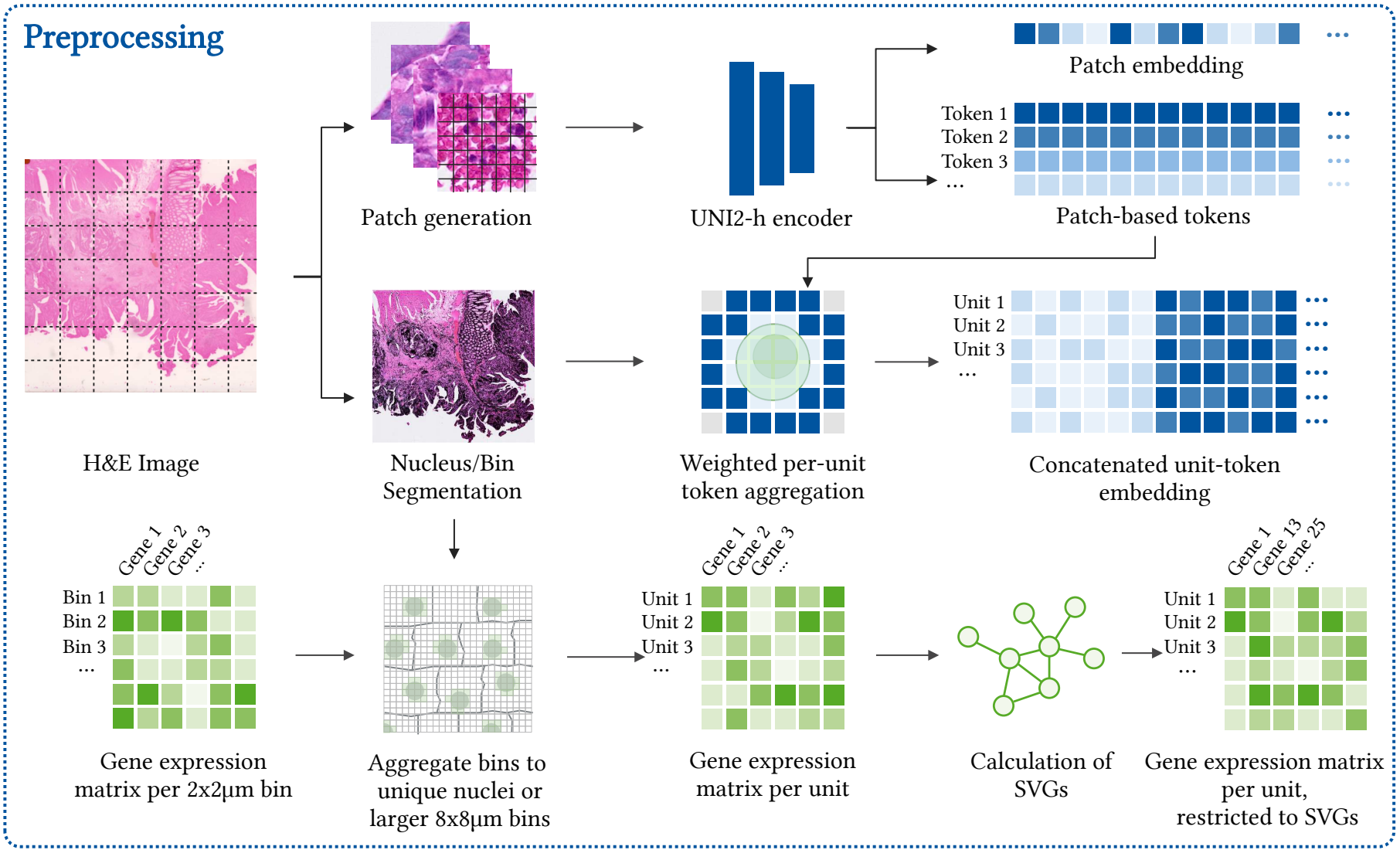

Preprocessing
Patch generation
UNI2-h encoder
Patch embedding
Token 1
Token 2
Token 3
...
Patch-based tokens
H&E Image
Nucleus/Bin Segmentation
Weighted per-unit token aggregation
Unit 1
Unit 2
Unit 3
...
Concatenated unit-token embedding
Gene 1
Gene 2
Gene 3
...
Bin 1
Bin 2
Bin 3
...
Gene expression matrix per 2x2µm bin
Aggregate bins to unique nuclei or larger 8x8µm bins
Gene expression matrix per unit
Calculation of SVGs
Gene 13
Gene 25
Gene expression matrix per unit, restricted to SVGs